

Challenge 6 – AI-Driven Cyclone & Coastal Disaster Early Warning System

Domain: Disaster Management

The Challenge

Gujarat's 1,600+ km coastline is highly vulnerable to cyclones (e.g., Tauktae, Biparjoy), yet coastal communities and fishermen often receive delayed or generalized warnings, leading to preventable loss of life, boats, and property.

Objective

Build Agentic AI solutions that provide hyperlocal cyclone tracking, fishermen safety alerts, and coordinated evacuation and relief planning for coastal districts.

Suggested AI Agents

- Cyclone Track & Intensity Prediction Agent
- Fishermen Safety Alert Agent
- Evacuation Route Planning Agent
- Relief Resource Coordination Agent
- Post-Disaster Damage Assessment Agent

Technology to be used – IBM Bob + IBM Granite LLM + IBM cloud